

Takshak Singhania

takshaksinghanian1@gmail.com | (+91) 9351595646 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [CodeChef](#)

SUMMARY

Software Engineering student at IIIT Bhopal with experience building web applications and full-stack projects using **JavaScript**, **TypeScript**, **React**, **Node.js**, and **REST APIs**. Experienced in **collaborative development**, **algorithmic problem solving**, **testing**, and **CI/CD** through multi-person projects and development teams. Strong foundation in **data structures**, **algorithms**, **operating systems**, and **object-oriented programming**.

EDUCATION

Indian Institute of Information Technology, Bhopal

Sept 2023 – Jun 2027

B.Tech. in Electronics and Communication Engineering

Relevant Coursework: **Data Structures & Algorithms**, **Operating Systems**, **Object-Oriented Programming**, **Database Management**

EXPERIENCE

Axios Development Club, IIIT Bhopal

Sep 2025 – Present

Web Development Lead | Part-Time

- Managed **6-member team** using Agile sprints; improved code quality through GitHub-based peer reviews and CI checks.
- Streamlined deployment workflows by automating build processes with **GitHub Actions** and Git version control.

PROJECTS

PayFlow | TypeScript, React, Node.js, PostgreSQL, Redis, BullMQ, Docker

[GitHub](#)

- Built a payment processing simulator with PostgreSQL transactions and row-level locking to keep payment updates consistent when multiple requests happen at the same time.
- Prevented duplicate payment requests by using request keys and database constraints to safely handle simultaneous submissions.
- Added background webhook processing with **Redis** and **BullMQ**, including automatic retries, secure request signatures, and **58 automated unit and integration tests**.

Vox | Next.js, TypeScript, Tailwind CSS, WebRTC, LiveKit, SSE

[GitHub](#)

- Built a multimodal AI platform combining **text chat**, **voice conversations**, and **image generation** in a single web application.
- Integrated **LiveKit WebRTC** for real-time voice communication and streaming responses for interactive conversations.
- Added AI-powered image generation and a responsive interface using **Next.js**, **TypeScript**, and **Tailwind CSS**, with API credentials kept on the server.

Sentosa QR Ordering System | React, TypeScript, Node.js, PostgreSQL, Prisma, Socket.IO, Razorpay

[GitHub](#)

- Built a mobile QR ordering system that lets customers browse menus, customize items, place orders, make payments, and call staff from their table.
- Added secure table sessions and server-side price calculation to prevent customers from accessing other tables or changing order prices.
- Added real-time kitchen updates and **2-role RBAC** for manager and staff access, with permissions checked across **37 endpoints**. Integrated **Razorpay payments** with secure signature verification and webhooks, backed by **63 automated tests** covering authentication, security, payments, and order flows.

TECHNICAL SKILLS

- Languages:** JavaScript (ES6+), TypeScript, SQL, C/C++, Java, Go
- Frontend:** React.js, HTML5, CSS3, Tailwind CSS, Material UI, Redux Toolkit, Responsive Design, DOM Manipulation
- Backend:** Node.js, Express.js, REST APIs, BullMQ, Asynchronous Queues, Full Stack Development, API Security
- Database & Caching:** PostgreSQL, Redis, MongoDB, Relational Schema Design, Query Optimization, Row-Level Locking
- Testing:** Jest, Supertest, Unit Testing, Integration Testing, Postman
- Tools & DevOps:** Docker, Docker Compose, Git, GitHub, Linux, CI/CD, GitHub Actions, Agile/Scrum, VS Code

ACHIEVEMENTS

- Successfully solved **300+ questions** on LeetCode, mastering advanced data structures and algorithmic concepts.
- Achieved a **3-star rating** on CodeChef, demonstrating strong problem-solving abilities and competitive programming skills.